

Matrix inequalities and norms

Dimension-independent Hlawka constants for Schatten norms

Difficulty: challenging **Importance:** interesting to the community**Status:** Partially resolved

Rating rationale: Sharp cancellation constants for entire Schatten classes require new control across the Schatten scale, a challenging norm-geometry question; dimension-independent control would inform perturbation estimates across the community.

Partial result — 2026-09-11

Partial result by Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent proof review: PASS for the stated partial result.**

The trace-norm endpoint has $C_1 = \infty$: a real 2×3 rank-one family has defect ratio asymptotic to $2/(3t)$ as $t \downarrow 0$. Zero-row padding gives the same failure on real 3×3 matrices.

Still open: The remaining finite Schatten exponents $1 < p < \infty$ are undetermined by this result; the known value $C_2 = 1$ is unchanged. The ratings apply to this surviving question.

Primary manuscript: [complete proof PDF](#), [standalone TeX](#), Theorem 1.1 and its proof; [authorship and scope](#). The [independent review](#) checks the full original argument and records its hash. The draft was AI-assisted; this is independent agent verification, not external human peer review or formal certification. [Submission record](#).

Problem statement

For each $1 \leq p < \infty$, determine the smallest $C_p \in [0, \infty]$ such that for every pair of positive integers N, M and every $X, Y, Z \in \mathbb{C}^{N \times M}$,

$$\|X\|_p + \|Y\|_p + \|Z\|_p - \|X + Y + Z\|_p \leq C_p (D_p(X, Y) + D_p(X, Z) + D_p(Y, Z)),$$

where $D_p(U, V) = \|U\|_p + \|V\|_p - \|U + V\|_p$ and $\|U\|_p = (\sum_j s_j(U)^p)^{1/p}$. The value $C_p = \infty$ means that no finite constant works uniformly over dimensions. The inequality must also hold when the parenthesized expression vanishes; a ratio definition must not discard such triples.

This asks whether the cancellation loss in a three-term matrix sum can be controlled sharply by the cancellation losses of its pairs. Such estimates concern the geometry of the matrix norms used in perturbation bounds and low-rank approximation.

References

1. K. M. R. Audenaert and F. Kittaneh, *Problems and Conjectures in Matrix and Operator Inequalities*, arXiv:1201.5232v3 (2012), §8.2, Problem 7, equation (48), and the following discussion. [Full text](#).
2. Audenaert and Kittaneh, published version in *Études opératorielles*, Banach Center Publications 112 (2017), §8.2, Problem 6, equation (48), p. 25. [Publisher PDF](#).

Status check (2026-09-10): the source gives $C_2 = 1$ and demonstrates nonexistence of a finite constant for the operator norm, while posing the Schatten constants generally. Current source records and searches for “Hlawka Schatten constant”, “Hlawka operators Audenaert Kittaneh”, and 2025/2026 found no determination of all C_p . Later determinantal Hlawka inequalities concern a different inequality and do not answer this norm-deficit question. The search is bounded; individual p -cases should be rechecked before treating them as new results.